HAN CHIANG HIGH SCHOOL, PENANG
TRIAL EXAMINATION 2026
ADVANCED MATHEMATICS II
(SE07)

PAPER 1 OBJECTIVE QUESTIONS

Class : S3
Duration : 60 minutes

INSTRUCTIONS TO CANDIDATES

1. This subject comprises of two papers:
   Paper 1 : Multiple-choice question (30%)
   Paper 2 : Subjective question (70%)
2. Complete Paper 1 within 60 minutes allocated.
3. Attempt all 15 questions in Paper 1. For each question, select the correct answer and blacken the square corresponding the same letter of the alphabet as a answer you have selected. Mark the answers with a 2B pencil on answer sheet 'O' (computer card)
4. Electronic calculator may be used.
5. This paper consists of 6 printed pages.

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Student's Name: _____ Student's Number: _____

Advanced Mathematics (II) Formula Sheet

I. Algebra

$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$

$\log_a (xy) = \log_a x + \log_a y$

$\log_a \dfrac{x}{y} = \log_a x - \log_a y$

$\log_a x^m = m \log_a x$

$a^{\log_a x} = x$

$\log_a x = \dfrac{\log_b x}{\log_b a}$

$(a+b)^n = \sum_{r=0}^{n} {}^nC_r \, a^{n-r} b^r$

$A^{-1} = \dfrac{1}{|A|} \operatorname{adj} A$

$\dfrac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}$

$\left[r(\cos\theta + i\sin\theta)\right]^n = r^n (\cos n\theta + i \sin n\theta)$

$(1+x)^n = 1 + \dfrac{n}{1!}x + \dfrac{n(n-1)}{2!}x^2 + \cdots + \dfrac{n(n-1)\cdots(n-r+1)}{r!}x^r + \cdots$

A.P. $a_n = a + (n-1)d$

$S_n = \dfrac{n}{2}\left[2a + (n-1)d\right]$

G.P. $a_n = ar^{n-1}$

$S_n = \dfrac{a(1-r^n)}{1-r}$

$S_\infty = \dfrac{a}{1-r}$

$\sum_{k=1}^{n} k = \dfrac{n(n+1)}{2}$

$\sum_{k=1}^{n} k^2 = \dfrac{n(n+1)(2n+1)}{6}$

$\sum_{k=1}^{n} k^3 = \left[\dfrac{n(n+1)}{2}\right]^2$

Accumulated value after n years $= p \times \left(1 + \dfrac{\frac{r}{m}}{100}\right)^{nm}$

II. Trigonometry

Arc length $= \theta r$

Area of sector $= \dfrac{1}{2}\theta r^2$

$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2\mathrm{R}$

$a^2 = b^2 + c^2 - 2bc\cos A$

Area of triangle $= \dfrac{1}{2}ab\sin C$

Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$,

$s = \dfrac{a+b+c}{2}$

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \operatorname{cosec}^2\theta$

$\sin(A \pm B) = \sin A\cos B \pm \cos A \sin B$

$\cos(A \pm B) = \cos A\cos B \mp \sin A \sin B$

$\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

$\sin 2A = 2\sin A\cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$= 2\cos^2 A - 1$

$= 1 - 2\sin^2 A$

$\tan 2A = \dfrac{2\tan A}{1 - \tan^2 A}$

$\sin A\cos B = \dfrac{\sin(A+B) + \sin(A-B)}{2}$

$\cos A\cos B = \dfrac{\cos(A+B) + \cos(A-B)}{2}$

$\sin A\sin B = \dfrac{\cos(A-B) - \cos(A+B)}{2}$

III. Analytic Geometry

Distance from a point to a line $d = \dfrac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$

Division of a line segment $\left(\dfrac{mx_2 + nx_1}{m+n}, \dfrac{my_2 + ny_1}{m+n} \right)$

Area of triangle $= \dfrac{1}{2} \left| (x_1y_2 + x_2y_3 + x_3y_1) - (x_2y_1 + x_3y_2 + x_1y_3) \right|$

Equation of a line $y - y_1 = m(x - x_1)$

Standard equation of a circle $(x-h)^2 + (y-k)^2 = r^2$

Translation $\begin{cases} x = x' + h \\ y = y' + k \end{cases}$

Rotation of axes $\begin{cases} x = x'\cos\theta - y'\sin\theta \\ y = x'\sin\theta + y'\cos\theta \end{cases}$

Parabola

Standard equation	$y^2 = 4ax$
Focus	$(a, 0)$
Directrix	$x + a = 0$

Ellipse

Standard equation	$\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$
Eccentricity	$e = \dfrac{\sqrt{a^2 - b^2}}{a}$
Foci	$(\pm ae, 0)$
Directrices	$x \pm \dfrac{a}{e} = 0$

Hyperbola

Standard equation	$\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$
Eccentricity	$e = \dfrac{\sqrt{a^2 + b^2}}{a}$
Foci	$(\pm ae, 0)$
Directrices	$x \pm \dfrac{a}{e} = 0$
Asymptotes	$y = \pm \dfrac{b}{a} x$

Inner product $\underline{a} \cdot \underline{b} = |\underline{a}||\underline{b}|\cos\theta$

Magnitude of the cross product $|\underline{a} \times \underline{b}| = |\underline{a}||\underline{b}|\sin\theta$

Projection of vector $\underline{a}$ onto vector $\underline{u}$ $\quad \text{proj}_{\underline{u}}\,\underline{a} = \dfrac{\underline{a} \cdot \underline{u}}{|\underline{u}|^2}\underline{u}$

IV. Statistics and Probability

Mean $\bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n}$

Mode $= L + \left(\dfrac{d_1}{d_1 + d_2} \right) C$

Median $m = L_m + \left(\dfrac{\frac{n}{2} - F_m}{f_m} \right) C_m$

Upper quartile $Q_3 = L_3 + \left(\dfrac{\frac{3n}{4} - F_3}{f_3} \right) C_3$

Lower quartile $Q_1 = L_1 + \left(\dfrac{\frac{n}{4} - F_1}{f_1} \right) C_1$

Interquartile range $IQR = Q_3 - Q_1$

Quartile deviation $Q.D. = \dfrac{Q_3 - Q_1}{2}$

Variance $\sigma^2 = \dfrac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n} = \dfrac{\sum_{i=1}^{n} x_i^2}{n} - \bar{x}^2$

Covariance $\sigma_{XY} = \dfrac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$

$= \dfrac{1}{n} \sum_{i=1}^{n} (x_i y_i) - \bar{x} \cdot \bar{y}$

Coefficient of correlation $r = \dfrac{\sigma_{XY}}{\sigma_X \sigma_Y}$

Equation of the regression line

$\hat{y} - \bar{y} = \dfrac{\sigma_{XY}}{\sigma_X^2} (x - \bar{x})$

Statistical index $I = \dfrac{Q_1}{Q_0} \times 100$

Composite index $= \dfrac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$

${}^{n}P_r = \dfrac{n!}{(n-r)!}$

${}^{n}C_r = \dfrac{n!}{(n-r)!r!}$

$P(E_k \mid A) = \dfrac{P(A \cap E_k)}{\sum_{i=1}^{n} P(A \mid E_i) P(E_i)}$

Random variable X

$E(X) = x_1 p_1 + x_2 p_2 + \cdots + x_k p_k$

$\sigma_X^2 = E(X^2) - (E(X))^2$

Binomial distribution $P(X = r) = {}^{n}C_r \, p^r (1-p)^{n-r}$

$\mu = np$

$\sigma = \sqrt{np(1-p)}$

Normal distribution $Z = \dfrac{X - \mu}{\sigma}$

Answer all questions.

1. Find the coordinates of the focus of the parabola $x^2 - 6x + 20y - 11 = 0$.

A. (-2, 1) B. (3,1) C. (3, -4) D. (3,6)

2. Find the coordinates of $\left(\sqrt{3}, 1\right)$ in the new coordinate system after a rotation of the coordinates axes through $\frac{\pi}{3}$

A. $\left(-\sqrt{3}, 1\right)$ B. $\left(-\sqrt{3}, -1\right)$ C. $\left(\sqrt{3}, -1\right)$ D. $\left(-1, \sqrt{3}\right)$

3. If $x^3 + 3x^2 + ax + 2$ is divisible by $x^2 + bx + 1$, find $a + 2b$.

A. 2 B. 3 C. 4 D. 5

4. Solve inequality $\left|\sqrt{4-x} - 3\right| < 5$

A. $-60 < x < 0$ B. $-60 < x < 4$ C. $-60 < x \leq 4$ D. $x > -60$

5. Given that $\int_3^4 f(x)\,dx = 21$, $\int_3^7 f(x)\,dx = -15$, find $\int_1^2 f(3x+1)\,dx$

A. -36 B. 36 C. -12 D. 12

6. Given that $\left(\sqrt{3} + i\right)^n$ is a purely imaginary number, find the possible value of n.

A. 7 B. 8 C. 9 D. 10

7. If the function $f : R \to R$ satisfies $f(6x+1) = 2x - 5$, find $f^{-1}(4x-1)$.

A. $8x + 9$ B. $10x + 11$ C. $12x + 13$ D. $14x + 15$

8. Given that $(7-i)(3+2i)^2 + a(5+7i) = b + 23i$, find the value of $a+3b$.

A. 12 B. 13 C. 14 D. 15

9. Find the equation of the circle centred at (-2, 1) that touches the line $3x - 4y - 5 = 0$.

A. $(x+2)^2 + (y-1)^2 = 9$ B. $(x+2)^2 + (y-1)^2 = 15$

C. $(x+2)^2 + (y-1)^2 = 45$ D. $(x+2)^2 + (y-1)^2 = 27$

10. Given that $g(x) = x^2 - 12x + 6$, $f(x) = x^8 + x^4 + a$ and x is factor of $g[f(x)]$.
Find the value of a.

A. $3 \pm \sqrt{2}$ B. $2 \pm \sqrt{3}$ C. $5 \pm 2\sqrt{3}$ D. $6 \pm \sqrt{30}$

11. Given that $f(x)=\begin{cases} 2x+4, & x<-1 \\ 2x^2, & -1\le x\le 2 \\ x+6, & x>2 \end{cases}$, if $f(x)=4$, find the value of x.

A. 10 B. 2 C. $\pm\sqrt{2}$ D. $\sqrt{2}$

12. On which of the following intervals is $f(x)=8x^4+4x^2-2$ increasing?

A. -2 B. 0 C. 20 D. 50

13. If the maximum distance between the two points on the ellipse $(x+3)^2+8y^2=k$ is 16, find the value of k.

A. 4 B. 8 C. 16 D. 64

14. The equation of the normal to the parabola $y^2-2x+2y+3=0$ at point (3, 1) is $Ax+By+C=0$, Find the value of $A+B+C$.

A. -2 B. -5 C. -3 D. -4

15. Find the value of $\int_0^1 x(3x-2)^7\,dx$

A. $-\frac{3}{4}$ B. $-\frac{4}{3}$ C. $-\frac{3}{5}$ D. $-\frac{5}{3}$